MTH 200 Chapter 2: Limits

2.3.1 & 2.3.2 Worksheet

Problems #1-8: Find the following limits. If the limit does not exist provide proper justification for stating that it does not.

1. $\lim\limits_{x \to 1.5^-} \dfrac{e^x}{2x-3}$

2. $\lim\limits_{x \to 2} \dfrac{\sqrt{4x+1}-3}{x-2}$

3. $\lim\limits_{x \to 0} \dfrac{\frac{1}{2}-\frac{1}{x+2}}{x}$

4. $\lim_{x\to-\infty} \frac{5+2x^3}{3x^3+2x-4}$

5. $\lim_{x\to\infty} \frac{2-5e^{2x}}{3e^{2x}}$

6. $\lim_{x\to\infty} \sqrt{x^2+x}-x$

7. $\lim_{x\to-\infty} \frac{x^3+3x+2}{4-5x^2+4x}$

8. $\lim_{x\to\infty} \frac{3+2\ln(x)}{5-6\ln(x)}$

9. Find all horizontal asymptotes on the graph of $f(x)=\frac{\sqrt{3x^4+x}}{x^2-8}$.